

**MODEL QUESTION PAPER**  
**QUANTITATIVE TECHNIQUES- SET I**

(Maximum marks: 75)

(Time: 3 hours)

**PART A**

**I. Answer all questions in one word or one sentence. Each question carries one mark.**

**(9 x 1 = 9 marks)**

|   |                                                                                                                                                |        |   |
|---|------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|
| 1 | The data which have already been collected by someone are called _____.                                                                        | M 1.04 | R |
| 2 | A branch of mathematics dealing with the collection, analysis, interpretation, and presentation of masses of numerical data is known as _____. | M 1.01 | R |
| 3 | _____ is the positional average of middle, which is why we need to arrange the data before we can perform any calculation.                     | M 2.02 | R |
| 4 | “A measure of central tendency is a single value which is used to represent an entire set of data.” True/False.                                | M 2.02 | A |
| 5 | _____ refers to the relationship between any two or more variables.                                                                            | M 3.01 | R |
| 6 | A relationship between two variables in which both variables move in the same direction is known as _____.                                     | M 3.02 | R |
| 7 | A period for which index number is determined is called as -----                                                                               | M 4.01 | R |
| 8 | Price relatives are a percentage ratio of current year price and _____.                                                                        | M 4.01 | R |
| 9 | Find Mode: 8, 5, 7, 10, 15, 21, 5, 7, 2, 5                                                                                                     | M 2.04 | A |

**PART B**

**II Answer any eight questions from the following. Each question carries 3 marks**

**(8 x 3 = 24marks)**

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |   |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|
| 1 | Enumerate the objectives of statistics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | M 1.01 | U |
| 2 | Point out the demerits of mean.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | M 2.03 | U |
| 3 | Explain the importance of Correlation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | M 3.01 | U |
| 4 | Outline the uses of Index Numbers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | M 4.01 | U |
| 5 | Describe the methods of Collecting Secondary data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | M 1.04 | U |
| 6 | Calculate Arithmetic Mean of the daily income of 5 families:<br>Rs. 10,90,85,103,11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | M 2.04 | A |
| 7 | Explain Karl Pearson's Co-efficient of correlation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | M 3.03 | U |
| 8 | For the data given below calculate Simple Index Number<br><div style="display: flex; justify-content: space-around; margin-top: 5px;"> <div>Commodities</div> <div>Price (2007)</div> <div>Price (2008)</div> </div> <div style="display: flex; justify-content: space-around; margin-top: 5px;"> <div>A</div> <div>5</div> <div>7</div> </div> <div style="display: flex; justify-content: space-around; margin-top: 5px;"> <div>B</div> <div>8</div> <div>9</div> </div> <div style="display: flex; justify-content: space-around; margin-top: 5px;"> <div>C</div> <div>12</div> <div>15</div> </div> <div style="display: flex; justify-content: space-around; margin-top: 5px;"> <div>D</div> <div>25</div> <div>24</div> </div> <div style="display: flex; justify-content: space-around; margin-top: 5px;"> <div>E</div> <div>3</div> <div>4</div> </div> | M 4.04 | A |
| 9 | Find range and its co efficient                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | M 2.07 | A |

|    |                                                                |        |   |
|----|----------------------------------------------------------------|--------|---|
|    | Weight : 5 8 10 12 25 30 38<br>No of children : 2 3 8 10 9 3 2 |        |   |
| 10 | Enumerate the merits of Mode.                                  | M 2.03 | U |

### PART C

**Answer all questions. Each question carries seven marks**

**(6x7=42 marks)**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|
| III  | State the meaning and Explain the various steps involved in statistical enquiry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | M 1.03 | U |
| IV   | OR<br>Describe the limitations of Statistics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | M 1.02 | U |
| V    | Compute Median<br>Class : 0-10 10-20 20-30 30-40 40-50 50-60 60-70<br>Frequency: 8 12 20 23 18 7 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M2.04  | A |
| VI   | OR<br>Enumerate the requisites of good average                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | M2.03  | U |
| VII  | Classify different types of Correlation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | M3.02  | U |
| VIII | OR<br>The ranking of 10 individuals at the start and at the finish of a course of a training are as follows:<br>Individuals: A B C D E F G H I J<br>Rank before: 1 6 3 9 5 2 7 10 8 4<br>Rank after : 6 8 3 2 7 10 5 9 4 1                                                                                                                                                                                                                                                                                                                                                                                                        | M3.04  | A |
| IX   | Summarize the problems in construction of Index Numbers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | M4.02  | U |
| X    | OR<br>From the following data construct index number under Fisher's Ideal Index Method.<br><br>Commodities                      Price                      Quantity<br>Base Year   Current Year   Base Year   Current Year<br>A                      5                      7                      3                      4<br>B                      8                      12                      7                      5<br>C                      15                      20                      6                      4<br>D                      6                      8                      3                      4 | M4.04  | A |
| XI   | Find Standard Deviation from the following data<br>Size : 0-2 2-4 4-6 6-8 8-10 10-12<br>Frequency: 2 4 6 4 2 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | M2.07  | A |
| XII  | OR<br>Calculate Weighted mean of the following data<br>Size : 5 10 15 20 25 30<br>Weight : 8 4 5 10 7 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | M2.05  | A |
| XIII | Calculate Karl Pearson's Coefficient of Correlation<br>X : 3 4 5 6 7<br>Y : 5 6 7 8 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | M3.04  | A |

|     |                                  |       |   |
|-----|----------------------------------|-------|---|
| XIV | OR<br>Explain Linear Regression. | M3.05 | U |
|-----|----------------------------------|-------|---|

### Question Wise Analysis

**Course : Quantitative Techniques**

| Q. No | Module Outcome | Cognitive Level | Marks | Time   |
|-------|----------------|-----------------|-------|--------|
| I.1   | M 1.04         | R               | 1     | 1 min  |
| I.2   | M 1.01         | R               | 1     | 1 min  |
| I.3   | M2.02          | R               | 1     | 1 min  |
| I.4   | M2.02          | A               | 1     | 1 min  |
| I.5   | M3.01          | R               | 1     | 1 min  |
| I.6   | M3.02          | R               | 1     | 1 min  |
| I.7   | M4.01          | R               | 1     | 1 min  |
| I.8   | M4.01          | R               | 1     | 1 min  |
| I.9   | M2.04          | A               | 1     | 1 min  |
| II.1  | M 1.01         | U               | 3     | 6 min  |
| II.2  | M 2.03         | U               | 3     | 6 min  |
| II.3  | M 3.01         | U               | 3     | 6 min  |
| II.4  | M 4.01         | U               | 3     | 6 min  |
| II.5  | M 1.04         | U               | 3     | 6 min  |
| II.6  | M2.04          | A               | 3     | 6 min  |
| II.7  | M 3.03         | U               | 3     | 6 min  |
| II.8  | M4.04          | A               | 3     | 6 min  |
| II.9  | M2.07          | A               | 3     | 6 min  |
| II.10 | M 2.03         | U               | 3     | 6 min  |
| III   | M 1.03         | U               | 7     | 20 min |
| IV    | M 1.02         | U               | 7     | 20 min |
| V     | M2.04          | A               | 7     | 20 min |

|      |        |   |   |        |
|------|--------|---|---|--------|
| VI   | M 2.03 | U | 7 | 20 min |
| VII  | M 3.02 | U | 7 | 20 min |
| VIII | M3.04  | A | 7 | 20 min |
| IX   | M 4.02 | U | 7 | 20 min |
| X    | M4.04  | A | 7 | 20 min |
| XI   | M2.07  | A | 7 | 20 min |
| XII  | M2.05  | A | 7 | 20 min |
| XIII | M3.04  | A | 7 | 20 min |
| XIV  | M3.05  | U | 7 | 20 min |

**Course : Quantitative Techniques**

### **Blue Print**

### **Mark Distribution**

| Module | hr / Module | Marks/Module<br>( $h_i / \sum H_i$ ) * 123 ( $\pm 5\%$ ) | Type of Questions |       |                 |       |                 |       |                 |       |
|--------|-------------|----------------------------------------------------------|-------------------|-------|-----------------|-------|-----------------|-------|-----------------|-------|
|        |             |                                                          | Part A            |       | Part B          |       | Part C          |       | Total           |       |
|        |             |                                                          | No of Questions   | Marks | No of Questions | Marks | No of Questions | Marks | No of Questions | Marks |
| 1      | 14          | 30                                                       | 2                 | 2     | 2               | 6     | 2               | 14    | 6               | 22    |
| 2      | 17          | 36                                                       | 3                 | 3     | 4               | 12    | 4               | 28    | 11              | 43    |
| 3      | 14          | 30                                                       | 2                 | 2     | 2               | 6     | 4               | 28    | 8               | 36    |
| 4      | 13          | 27                                                       | 2                 | 2     | 2               | 6     | 2               | 14    | 6               | 22    |
| Total  | 58          | 123                                                      | 9                 | 9     | 10              | 30    | 12              | 84    | 31              | 123   |

**Blue Print**

**Cognitive Level Mark Distribution**

| <b>Cognitive Level</b> | <b>Marks</b> | <b>% of Marks</b> |
|------------------------|--------------|-------------------|
| Remembering            | 7            | 6                 |
| Understanding          | 63           | 51                |
| Applying               | 53           | 43                |
| Analysing              |              |                   |
| Evaluating             |              |                   |
| Creating               |              |                   |
| <b>Total</b>           | <b>123</b>   | <b>100</b>        |

**SCHEME OF VALUATION**  
**(Scoring Indicators)- Set I**  
**Quantitative Techniques**

| <b>Qn.<br/>No</b> | <b>Scoring Indicators</b> | <b>Split<br/>up<br/>Score</b> | <b>Sub<br/>Total</b> | <b>Total</b> |
|-------------------|---------------------------|-------------------------------|----------------------|--------------|
| <b>I</b>          | <b>PART A</b>             |                               |                      |              |
| 1                 | Secondary data            | 1                             | 1                    | 1            |
| 2                 | Statistics                | 1                             | 1                    | 1            |
| 3                 | Median                    | 1                             | 1                    | 1            |
| 4                 | True                      | 1                             | 1                    | 1            |
| 5                 | Correlation               | 1                             | 1                    | 1            |
| 6                 | Positive Correlation      | 1                             | 1                    | 1            |
| 7                 | Current year              | 1                             | 1                    | 1            |
| 8                 | Base year price           | 1                             | 1                    | 1            |
| 9                 | Mode is 5.                | 1                             | 1                    | 1            |

| II | PART B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |   |   |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---|
| 1  | <b>Objectives of Statistics</b><br><br>1. To collect all the statistical facts related to a particular problem.<br><br>2.To compare the past and present statistical facts.<br><br>3.To find out the reason for changes.<br><br>4.To investigate those reason.<br><br>5.To focus on problem for which the investigation has been put on.<br><br>6.To offer solution.<br><br>(Any 3 points mentioned above)                                                                                         | 3X1 | 3 | 3 |
| 2  | <b>Demerits of Arithmetic Mean</b><br><br>1. Arithmetic Mean sometimes gives most absurd result. Eg: Average children in a family as 3.5 or 2.8<br>2. It cannot be used in the study of qualitative phenomena like intelligence, beauty etc.<br>3. It is affected by extreme values.<br>4. It is not suitable for averaging ratios and percentages.<br>5. If a single observation is missing or lost, mean cannot be calculated.<br><br>(Any 3 points mentioned above)                             | 3X1 | 3 | 3 |
| 3  | <b>Importance of Correlation</b><br><br>1.It is used to reducing the range of uncertainty<br><br>2.It is used in presenting the average relationship between any two variables through a single value of co-efficient of correlation.<br><br>3.In economics, it is used in understanding the economic behaviour.<br><br>4.In business, it is used to estimate the cost of sales, sales price etc.<br><br>5.In science, it is used in measuring the progress.<br><br>(Any 3 points mentioned above) | 3X1 | 3 | 3 |

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                           |       |   |    |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---|----|---|----|--|----|--|-----|--|-----------|---|-------------------|--|------------------------------------------------------------------------------------|---|---|---|---|---|
| 4                 | <p><b>Uses of Index Numbers</b></p> <p>1. Help in studying trend:</p> <p>2. Help in policy formulation:</p> <p>3. Help in measuring the purchasing power of money:</p> <p>4. Act as economic barometers:</p> <p>5 . Help in deflating various values:</p> <p>(Any 3 points mentioned above)</p>                                                                                                                           | 3X1   | 3 | 3  |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |
| 5                 | <p><b>Methods of collecting Secondary Data</b></p> <p>1. Published Sources:</p> <p>2. Unpublished Sources</p>                                                                                                                                                                                                                                                                                                             | 1.5X2 | 3 | 3  |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |
| 6                 | <p>Calculation of Arithmetic Mean</p> <table> <tr> <td>X</td> <td>1</td> </tr> <tr> <td>10</td> <td>1</td> </tr> <tr> <td>90</td> <td></td> </tr> <tr> <td>85</td> <td></td> </tr> <tr> <td>103</td> <td></td> </tr> <tr> <td><u>11</u></td> <td>1</td> </tr> <tr> <td><b><u>299</u></b></td> <td></td> </tr> </table> <p>Arithmetic Mean = <math>\frac{\sum x}{N}</math></p> <p><math>\sum x = 299, \quad N=5</math></p> | X     | 1 | 10 | 1 | 90 |  | 85 |  | 103 |  | <u>11</u> | 1 | <b><u>299</u></b> |  | <table> <tr> <td>1</td> </tr> <tr> <td>1</td> </tr> <tr> <td>1</td> </tr> </table> | 1 | 1 | 1 | 3 | 3 |
| X                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                         |       |   |    |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |
| 10                | 1                                                                                                                                                                                                                                                                                                                                                                                                                         |       |   |    |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |
| 90                |                                                                                                                                                                                                                                                                                                                                                                                                                           |       |   |    |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |
| 85                |                                                                                                                                                                                                                                                                                                                                                                                                                           |       |   |    |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |
| 103               |                                                                                                                                                                                                                                                                                                                                                                                                                           |       |   |    |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |
| <u>11</u>         | 1                                                                                                                                                                                                                                                                                                                                                                                                                         |       |   |    |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |
| <b><u>299</u></b> |                                                                                                                                                                                                                                                                                                                                                                                                                           |       |   |    |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |
| 1                 |                                                                                                                                                                                                                                                                                                                                                                                                                           |       |   |    |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |
| 1                 |                                                                                                                                                                                                                                                                                                                                                                                                                           |       |   |    |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |
| 1                 |                                                                                                                                                                                                                                                                                                                                                                                                                           |       |   |    |   |    |  |    |  |     |  |           |   |                   |  |                                                                                    |   |   |   |   |   |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |    |    |   |   |   |   |   |   |   |    |    |   |    |    |   |           |  |           |           |  |                               |   |   |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----|----|---|---|---|---|---|---|---|----|----|---|----|----|---|-----------|--|-----------|-----------|--|-------------------------------|---|---|
|             | <div>Mean = <math>\frac{299}{5}</math>= <u><b>59.8</b></u></div> <div>5</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |    |    |   |   |   |   |   |   |   |    |    |   |    |    |   |           |  |           |           |  |                               |   |   |
| 7           | <div><b>Karl Pearson's Coefficient of Correlation (Simple Correlation)</b></div> <div>The Karl Pearson's method popularly known as Pearsonian Co-efficient of correlation is most widely used in practice. The Co-efficient of correlation is the numeral value which shows the degree of relationship between two variables and is represented by the symbol 'r'. He asserts that the value of Co-efficient of correlation must be in between +1 and -1. His interpretation of the co efficient values or r are as follows:</div> <div>When r = 1, it is indicative of perfect positive correlation</div> <div>When r = -1, it is indicative of perfect negative correlation</div> <div>When r = &gt;0 but &lt; 1, it is a case of imperfect positive correlation</div> <div>When r = &lt; 0 but &gt; -1, it is a case of imperfect negative correlation</div> <div>When r = 0, it indicates no correlation</div> | 3X1         | 3  | 3  |   |   |   |   |   |   |   |    |    |   |    |    |   |           |  |           |           |  |                               |   |   |
| 8           | <div><b>Calculation of Simple Index Number</b></div> <div><table><tr><td>Commodities</td><td>P0</td><td>P1</td></tr><tr><td>A</td><td>5</td><td>7</td></tr><tr><td>B</td><td>8</td><td>9</td></tr><tr><td>C</td><td>12</td><td>15</td></tr><tr><td>D</td><td>25</td><td>24</td></tr><tr><td>E</td><td><u>34</u></td><td></td></tr><tr><td><b>53</b></td><td><b>59</b></td><td></td></tr></table></div> <div><b>Simple Index Number (P01)</b> = <math>\frac{\sum P1}{\sum P0} \times 100</math></div> <div>= <math>\frac{59}{53} \times 100</math></div> <div>= <u><b>111.32</b></u></div>                                                                                                                                                                                                                                                                                                                          | Commodities | P0 | P1 | A | 5 | 7 | B | 8 | 9 | C | 12 | 15 | D | 25 | 24 | E | <u>34</u> |  | <b>53</b> | <b>59</b> |  | <div>1.5</div> <div>1.5</div> | 3 | 3 |
| Commodities | P0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | P1          |    |    |   |   |   |   |   |   |   |    |    |   |    |    |   |           |  |           |           |  |                               |   |   |
| A           | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7           |    |    |   |   |   |   |   |   |   |    |    |   |    |    |   |           |  |           |           |  |                               |   |   |
| B           | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9           |    |    |   |   |   |   |   |   |   |    |    |   |    |    |   |           |  |           |           |  |                               |   |   |
| C           | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 15          |    |    |   |   |   |   |   |   |   |    |    |   |    |    |   |           |  |           |           |  |                               |   |   |
| D           | 25                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 24          |    |    |   |   |   |   |   |   |   |    |    |   |    |    |   |           |  |           |           |  |                               |   |   |
| E           | <u>34</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |    |    |   |   |   |   |   |   |   |    |    |   |    |    |   |           |  |           |           |  |                               |   |   |
| <b>53</b>   | <b>59</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |    |    |   |   |   |   |   |   |   |    |    |   |    |    |   |           |  |           |           |  |                               |   |   |
| 9           | <div><b>Range = H – L</b></div> <div>= 38 – 5</div> <div>= <u><b>33</b></u></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1.5         | 3  | 3  |   |   |   |   |   |   |   |    |    |   |    |    |   |           |  |           |           |  |                               |   |   |



|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                             |     |   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----|---|
|     | <p><b>Co efficient of Range = <math>\frac{H - L}{H + L}</math></b></p> <p><math>= \frac{38 - 5}{38 + 5}</math></p> <p><math>= \frac{33}{43} = .77</math></p>                                                                                                                                                                                                                                                                                                                                                                                                     | 1.5                         |     |   |
| 10  | <p><b>Merits of Mode</b></p> <ol style="list-style-type: none"> <li>1. Mode is very simple among Measures of central tendency.</li> <li>2. It gives the most representative value of a series.</li> <li>3. It is not affected by the extreme value of a series.</li> <li>4. It can be determined graphically also.</li> <li>5. Mode is the value which occurs most frequently.</li> <li>6. It is commonly understood and easily calculated</li> </ol> <p>(Any 3 points mentioned above)</p>                                                                      | 3X1                         | 3   | 3 |
|     | <b>PART C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                             |     |   |
| III | <p><b>Statistical Investigation</b> means investigation carried out by any agency wherein relevant data are collected and they are analysed with the help of different statistical methods.</p> <p><b>Stages in Statistical Investigation ( Statistical Enquiry)</b></p> <ol style="list-style-type: none"> <li>1. Planning the enquiry:</li> <li>2. Collection of data:</li> <li>3. Organisation of data:</li> <li>4. Presentation of the organised data:</li> <li>5. Analysing the presented data:</li> <li>6. Interpretation of the collected data</li> </ol> | <div>1</div> <div>6X1</div> | 1+6 | 7 |

## Limitations of Statistics

1. Statistics studies only numerical facts:
2. Statistics does not study individual cases:
3. Statistical results are true only on an average:
4. Statistics does not reveal the entire story of the problem:
5. Statistics is only one of the methods of studying a problem:
6. Statistics can be misused:
7. Only means and not a solution:
8. Statistical laws are not correct in the short period:

(Any 7 points mentioned above)

7X1

7

7

### Computation of Median

| Class | F            | Cf |
|-------|--------------|----|
| 0-10  | 8            | 8  |
| 10-20 | 12           | 20 |
| 20-30 | 20           | 40 |
| 30-40 | 23           | 63 |
| 40-50 | 18           | 81 |
| 50-60 | 7            | 88 |
| 60-70 | 2            | 90 |
|       | <b>N =90</b> |    |

3

$$\text{Median} = L1 + \frac{(N/2 - cf)}{f} \times i$$

$N/2 = 90/2 = 45$ ,  
 Median class 30-40  
 $f = 23$   
 $Cf = 40$

$$\begin{aligned} \text{Median} &= 30 + \frac{45-40}{23} \times 10 \\ &= 30 + \frac{5}{23} \times 10 \\ &= 30 + 2.17 \\ &= \mathbf{32.17} \end{aligned}$$

4

7

7

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |   |   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---|
| VI | <p><b>Requisites of good average</b></p> <ol style="list-style-type: none"> <li>1. It should be clearly defined:</li> <li>2. Its definition should be in the form of mathematical formula:</li> <li>3. It should be simple to understand:</li> <li>4. It should be simple to calculate:</li> <li>5. It should be based on all the observation of the series:</li> <li>6. It should be capable of further algebraic treatment:</li> <li>7. It should be capable of further statistical computation:</li> </ol> | 7X1 | 7 | 7 |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---|

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |                |            |                |   |   |    |    |   |   |    |   |   |   |   |   |   |   |   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------|------------|----------------|---|---|----|----|---|---|----|---|---|---|---|---|---|---|---|
| VII  | <p><b>Types of Correlation</b></p> <p><b>1. Simple, Partial and Multiple Correlation</b></p> <p>When the relationship between any two variables only studied, it is a case of simple correlation.</p> <p>When the relationship between any two out of three or more variables is studied, ignoring the effect of the other related variables, it is a case of partial correlation.</p> <p>When the relationship between three or more variables is studied simultaneously, it is a case of multiple correlation.</p> <p><b>2. Positive and Negative Correlation</b></p> <p>A <b>positive correlation</b> is one where values of both the variables under study move in the same direction. i.e, with an increase in the values of one variable, the values of other variable increases and with a decrease in the values of one variable, the values of other variable also decreases.</p> <p>A <b>negative correlation</b> is one where both the variables under study move in the opposite direction. i.e, an increase in the values of one variable is followed by a decrease in the values of the other, and a decrease in the values of one variable is followed by an increase in the values of the other variable.</p> <p><b>3. Linear and Non- linear Correlation</b></p> <p>When the amount of change in one variable leads to a constant ratio of change in the other variable, correlation is said to be linear.</p> <p>Correlation is said to be non-linear, when the amount of change in one variable is not in constant ratio to the change in the other variable.</p> | 7X1        | 7              | 7          |                |   |   |    |    |   |   |    |   |   |   |   |   |   |   |   |
| VIII | <table><tr><td>R1</td><td>R2</td><td>D ( R1-R2)</td><td>D<sup>2</sup></td></tr><tr><td>1</td><td>6</td><td>-5</td><td>25</td></tr><tr><td>6</td><td>8</td><td>-2</td><td>4</td></tr><tr><td>3</td><td>3</td><td>0</td><td>0</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | R1         | R2             | D ( R1-R2) | D <sup>2</sup> | 1 | 6 | -5 | 25 | 6 | 8 | -2 | 4 | 3 | 3 | 0 | 0 | 4 | 7 | 7 |
| R1   | R2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | D ( R1-R2) | D <sup>2</sup> |            |                |   |   |    |    |   |   |    |   |   |   |   |   |   |   |   |
| 1    | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | -5         | 25             |            |                |   |   |    |    |   |   |    |   |   |   |   |   |   |   |   |
| 6    | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | -2         | 4              |            |                |   |   |    |    |   |   |    |   |   |   |   |   |   |   |   |
| 3    | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0          | 0              |            |                |   |   |    |    |   |   |    |   |   |   |   |   |   |   |   |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <div> <div> <div>9</div> <div>2</div> <div>7</div> <div>49</div> </div> <div> <div>5</div> <div>7</div> <div>-2</div> <div>4</div> </div> <div> <div>2</div> <div>10</div> <div>-8</div> <div>64</div> </div> <div> <div>7</div> <div>5</div> <div>2</div> <div>4</div> </div> <div> <div>10</div> <div>9</div> <div>1</div> <div>1</div> </div> <div> <div>8</div> <div>4</div> <div>4</div> <div>16</div> </div> <div> <div>4</div> <div>1</div> <div>3</div> <div><u>9</u></div> </div> <div> <div>176</div> </div> <div> <math display="block">R = 1 - \frac{6 \sum D^2}{N^3 - N}</math> <math display="block">= 1 - \frac{6 \times 176}{1000 - 10}</math> <math display="block">= 1 - \frac{1056}{990}</math> <math display="block">= 1 - 1.07 = \underline{\underline{-.07}}</math> </div> </div> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |                |                |   |    |                |                |   |   |    |   |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |    |    |     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------|----------------|---|----|----------------|----------------|---|---|----|---|----|---|---|----|----|----|---|---|----|----|----|---|---|----|----|----|---|---|----|----|----|----|----|-----|
|      | <div><math>\Sigma w</math></div> <div><math>= 710/40</math></div> <div><math>= \underline{17.75}</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                |                |   |    |                |                |   |   |    |   |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |    |    |     |
| XIII | Karl Pearson’s Coefficient of Correlation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3   | 7              | 7              |   |    |                |                |   |   |    |   |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |    |    |     |
|      | <table><tr><td>X</td><td>y</td><td>Xy</td><td>x<sup>2</sup></td><td>y<sup>2</sup></td></tr><tr><td>3</td><td>5</td><td>15</td><td>9</td><td>25</td></tr><tr><td>4</td><td>6</td><td>24</td><td>16</td><td>36</td></tr><tr><td>5</td><td>7</td><td>35</td><td>25</td><td>49</td></tr><tr><td>6</td><td>8</td><td>48</td><td>36</td><td>64</td></tr><tr><td>7</td><td>9</td><td>63</td><td>49</td><td>81</td></tr><tr><td>25</td><td>35</td><td>185</td><td>135</td><td>255</td></tr></table> <div><math display="block">r = \frac{N \Sigma xy - (\Sigma x) (\Sigma y)}{\sqrt{N \Sigma x^2 - (\Sigma x)^2} \sqrt{N \Sigma y^2 - (\Sigma y)^2}}</math><math display="block">= \frac{5 \times 185 - 25 \times 35}{\sqrt{5 \times 135 - (25)^2} \times \sqrt{5 \times 255 - (35)^2}}</math><math display="block">= \frac{925 - 875}{\sqrt{675 - 625} \times \sqrt{1275 - 1255}}</math><math display="block">= \frac{50}{\sqrt{50} \times \sqrt{50}}</math><math display="block">= \underline{1}</math><div>As the Coefficient of Correlation is 1, it indicates perfect positive correlation.</div></div> | X   |                |                | y | Xy | x <sup>2</sup> | y <sup>2</sup> | 3 | 5 | 15 | 9 | 25 | 4 | 6 | 24 | 16 | 36 | 5 | 7 | 35 | 25 | 49 | 6 | 8 | 48 | 36 | 64 | 7 | 9 | 63 | 49 | 81 | 25 | 35 | 185 |
| X    | y                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Xy  | x <sup>2</sup> | y <sup>2</sup> |   |    |                |                |   |   |    |   |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |    |    |     |
| 3    | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 15  | 9              | 25             |   |    |                |                |   |   |    |   |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |    |    |     |
| 4    | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 24  | 16             | 36             |   |    |                |                |   |   |    |   |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |    |    |     |
| 5    | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 35  | 25             | 49             |   |    |                |                |   |   |    |   |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |    |    |     |
| 6    | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 48  | 36             | 64             |   |    |                |                |   |   |    |   |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |    |    |     |
| 7    | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 63  | 49             | 81             |   |    |                |                |   |   |    |   |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |    |    |     |
| 25   | 35                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 185 | 135            | 255            |   |    |                |                |   |   |    |   |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |   |   |    |    |    |    |    |     |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| XIV | <p><i>Linear regression</i> quantifies the relationship between one or more <i>predictor variable(s)</i> and one <i>outcome variable</i>. Linear regression is commonly used for predictive analysis and modeling. For example, it can be used to quantify the relative impacts of age, gender, and diet (the predictor variables) on height (the outcome variable). Linear regression is also known as <i>multiple regression</i>, <i>multivariate regression</i>, <i>ordinary least squares (OLS)</i>, and <i>regression</i>. This post will show you examples of linear regression, including an example of <i>simple linear regression</i> and an example of <i>multiple linear regression</i></p> <p>Linear regression is used for finding linear relationship between target and one or more predictors. There are two types of linear regression- Simple and Multiple.</p> <p>Simple Linear Regression</p> <p>Simple linear regression is useful for finding relationship between two continuous variables. One is predictor or independent variable and other is response or dependent variable. It looks for statistical relationship but not deterministic relationship. Relationship between two variables is said to be deterministic if one variable can be accurately expressed by the other. For example, using temperature in degree Celsius it is possible to accurately predict Fahrenheit. Statistical relationship is not accurate in determining relationship between two variables. For example, relationship between height and weight.</p> | 7 | 7 | 7 |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|

## MODEL QUESTION PAPER

### QUANTITATIVE TECHNIQUES- SET II

(Maximum marks: 75)

(Time: 3 hours)

#### PART A



**I. Answer all questions in one word or one sentence. Each question carries one mark.**

**(9 x 1 = 9 marks)**

|   |                                                                                                               |        |   |
|---|---------------------------------------------------------------------------------------------------------------|--------|---|
| 1 | The first hand and unorganized form of data is called _____.                                                  | M 1.04 | R |
| 2 | _____ is the process of arranging the data in groups or classes according to resemblances and similarities.   | M 1.05 | R |
| 3 | The _____ is the number that appears most frequently in a set.                                                | M 2.02 | R |
| 4 | Find out Range<br>Values: 25, 32, 85, 32, 42, 10, 20, 18, 28                                                  | M 2.07 | A |
| 5 | To measure ranked variables the _____ correlation coefficient is used.                                        | M 3.03 | R |
| 6 | A relationship between two variables in which both variables move in the opposite direction is known as _____ | M 3.02 | R |
| 7 | _____ are used for comparison of changes in the prices or quantities of different periods of time.            | M 4.01 | R |
| 8 | Questionnaire survey method is used to collect _____.                                                         | M 1.04 | R |
| 9 | The geometric mean of Laspeyre's and Paasche's price indices is also known as _____.                          | M 4.03 | R |

**PART B**

**II Answer any eight questions from the following. Each question carries 3 marks**

**(8 x 3 = 24marks)**

| 1           | Enumerate the essentials of good tabulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M 1.04       | U         |              |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------|--------------|----------|--|--|-----------|--------------|-----------|--------------|---|---|---|----|---|---|---|---|---|---|---|---|---|----|----|---|---|---|---|---|--------|---|
| 2           | Enumerate the merits of Arithmetic Mean.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | M 2.03       | U         |              |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
| 3           | Explain Spearman’s Rank Correlation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | M 3.04       | U         |              |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
| 4           | From the following data construct index number under Laspeyre’s Method. <table><tr><th>Commodities</th><th colspan="2">Price</th><th colspan="2">Quantity</th></tr><tr><td></td><th>Base Year</th><th>Current Year</th><th>Base Year</th><th>Current Year</th></tr><tr><td>A</td><td>4</td><td>7</td><td>10</td><td>8</td></tr><tr><td>B</td><td>5</td><td>9</td><td>8</td><td>6</td></tr><tr><td>C</td><td>6</td><td>8</td><td>15</td><td>12</td></tr><tr><td>D</td><td>2</td><td>2</td><td>5</td><td>6</td></tr></table> | Commodities  | Price     |              | Quantity |  |  | Base Year | Current Year | Base Year | Current Year | A | 4 | 7 | 10 | 8 | B | 5 | 9 | 8 | 6 | C | 6 | 8 | 15 | 12 | D | 2 | 2 | 5 | 6 | M 4.04 | A |
| Commodities | Price                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              | Quantity  |              |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
|             | Base Year                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Current Year | Base Year | Current Year |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
| A           | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 7            | 10        | 8            |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
| B           | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 9            | 8         | 6            |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
| C           | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 8            | 15        | 12           |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
| D           | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2            | 5         | 6            |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
| 5           | Describe the objectives of Classification.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | M 1.05       | U         |              |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
| 6           | Calculate median from the following<br>38, 24, 45, 50, 85, 60, 95, 40, 56, 63                                                                                                                                                                                                                                                                                                                                                                                                                                              | M 2.04       | A         |              |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
| 7           | Enumerate the importance of Correlation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | M 3.01       | U         |              |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
| 8           | Explain Simple aggregative method & Simple average relative method.                                                                                                                                                                                                                                                                                                                                                                                                                                                        | M 4.03       | U         |              |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
| 9           | Find mode from the values<br>40, 25, 60, 35, 81, 75, 90, 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M 2.04       | A         |              |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |
| 10          | Point out the demerits of Median.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | M 2.03       | U         |              |          |  |  |           |              |           |              |   |   |   |    |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |        |   |

**PART C**

**Answer all questions. Each question carries seven marks**

**(6x7=42 marks)**

|     |                                                                    |       |   |
|-----|--------------------------------------------------------------------|-------|---|
| III | Explain the methods of collecting Primary data and Secondary data. | M1.04 | U |
| OR  |                                                                    |       |   |
| IV  | Describe the functions of Statistics                               | M1.02 | U |

|                |                                                                                                                                                                                                                                                                                                                                        |       |       |       |       |       |       |       |       |       |                |   |    |    |    |    |    |    |   |        |   |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------------|---|----|----|----|----|----|----|---|--------|---|
| V              | Find Arithmetic Mean<br><div><div>Marks</div><div>No. of students</div><div>Less than 105</div><div>Less than 2017</div><div>Less than 3031</div><div>Less than 4041</div><div>Less than 5049</div></div> <div>OR</div>                                                                                                                | M2.04 | A     |       |       |       |       |       |       |       |                |   |    |    |    |    |    |    |   |        |   |
| VI             | Calculate Weighted mean of the following data<br>Size : 60 75 63 60 55<br>Weight :1 1 2 3 3                                                                                                                                                                                                                                            | M2.05 | A     |       |       |       |       |       |       |       |                |   |    |    |    |    |    |    |   |        |   |
| VII            | Calculate Karl Pearson’s Coefficient of Correlation<br>X :2 3 4 5 6 7 8<br>Y : 4 5 6 12 95 4                                                                                                                                                                                                                                           | M3.03 | A     |       |       |       |       |       |       |       |                |   |    |    |    |    |    |    |   |        |   |
| VIII           | Classify different types of Correlation.                                                                                                                                                                                                                                                                                               | M3.02 | U     |       |       |       |       |       |       |       |                |   |    |    |    |    |    |    |   |        |   |
| IX             | From the following data construct index number under Dorbish and Bowley Method.<br><div><div>Commodities</div><div>Price</div><div>Quantity</div><div>Base YearCurrent YearBase YearCurrent</div><div>Year</div><div>A5734</div><div>B81275</div><div>C152064</div><div>D6834</div></div> <div>OR</div>                                | M4.04 | A     |       |       |       |       |       |       |       |                |   |    |    |    |    |    |    |   |        |   |
| X              | Explain Weighted aggregative method &Weighted average relative method.                                                                                                                                                                                                                                                                 | M4.03 | U     |       |       |       |       |       |       |       |                |   |    |    |    |    |    |    |   |        |   |
| XI             | Calculate Median from the following data<br><table><tr><td>Marks</td><td>11-15</td><td>16-20</td><td>21-25</td><td>26-30</td><td>31-35</td><td>36-40</td><td>41-45</td><td>46-50</td></tr><tr><td>No of students</td><td>7</td><td>10</td><td>13</td><td>26</td><td>35</td><td>22</td><td>11</td><td>5</td></tr></table> <div>OR</div> | Marks | 11-15 | 16-20 | 21-25 | 26-30 | 31-35 | 36-40 | 41-45 | 46-50 | No of students | 7 | 10 | 13 | 26 | 35 | 22 | 11 | 5 | M 2.04 | A |
| Marks          | 11-15                                                                                                                                                                                                                                                                                                                                  | 16-20 | 21-25 | 26-30 | 31-35 | 36-40 | 41-45 | 46-50 |       |       |                |   |    |    |    |    |    |    |   |        |   |
| No of students | 7                                                                                                                                                                                                                                                                                                                                      | 10    | 13    | 26    | 35    | 22    | 11    | 5     |       |       |                |   |    |    |    |    |    |    |   |        |   |
| XII            | State the meaning and importance of Measures of Central Tendency.                                                                                                                                                                                                                                                                      | M2.01 | U     |       |       |       |       |       |       |       |                |   |    |    |    |    |    |    |   |        |   |
| XIII           | Find the Rank Correlation Co-efficient<br>X :17 13 15 16 6 11 14 9 7 12<br>Y : 36 46 35 24 12 18 27 22 2 8                                                                                                                                                                                                                             | M3.04 | A     |       |       |       |       |       |       |       |                |   |    |    |    |    |    |    |   |        |   |
| XIV            | From the following data of values of X and Y, find the regression equation of Y on X.                                                                                                                                                                                                                                                  | M3.05 | A     |       |       |       |       |       |       |       |                |   |    |    |    |    |    |    |   |        |   |

|  |       |   |   |   |   |  |  |
|--|-------|---|---|---|---|--|--|
|  | X : 2 | 3 | 4 | 5 | 6 |  |  |
|  | Y : 3 | 5 | 4 | 8 | 9 |  |  |

### Question Wise Analysis

**Course : Quantitative Techniques**

| Q. No | Module Outcome | Cognitive Level | Marks | Time   |
|-------|----------------|-----------------|-------|--------|
| I.1   | M 1.04         | R               | 1     | 1 min  |
| I.2   | M 1.05         | R               | 1     | 1 min  |
| I.3   | M2.02          | R               | 1     | 1 min  |
| I.4   | M2.07          | A               | 1     | 1 min  |
| I.5   | M3.03          | R               | 1     | 1 min  |
| I.6   | M3.02          | R               | 1     | 1 min  |
| I.7   | M4.01          | R               | 1     | 1 min  |
| I.8   | M1.04          | R               | 1     | 1 min  |
| I.9   | M4.03          | R               | 1     | 1 min  |
| II.1  | M 1.04         | U               | 3     | 6 min  |
| II.2  | M 2.03         | U               | 3     | 6 min  |
| II.3  | M 3.04         | U               | 3     | 6 min  |
| II.4  | M 4.04         | A               | 3     | 6 min  |
| II.5  | M 1.05         | U               | 3     | 6 min  |
| II.6  | M2.04          | A               | 3     | 6 min  |
| II.7  | M 3.01         | U               | 3     | 6 min  |
| II.8  | M4.03          | U               | 3     | 6 min  |
| II.9  | M2.04          | A               | 3     | 6 min  |
| II.10 | M 2.03         | U               | 3     | 6 min  |
| III   | M 1.04         | U               | 7     | 20 min |
| IV    | M 1.02         | U               | 7     | 20 min |

|      |        |   |   |        |
|------|--------|---|---|--------|
| V    | M2.04  | A | 7 | 20 min |
| VI   | M 2.05 | A | 7 | 20 min |
| VII  | M 3.03 | A | 7 | 20 min |
| VIII | M3.02  | U | 7 | 20 min |
| IX   | M 4.04 | A | 7 | 20 min |
| X    | M4.03  | U | 7 | 20 min |
| XI   | M2.04  | A | 7 | 20 min |
| XII  | M2.01  | U | 7 | 20 min |
| XIII | M3.04  | A | 7 | 20 min |
| XIV  | M3.05  | A | 7 | 20 min |

**Course : Quantitative Techniques**

**Blue Print**

**Mark Distribution**

| Module | hr / Module | Marks/Module<br>( $h_i / \sum H_i$ ) * 123 ( $\pm 5\%$ ) | Type of Questions |       |                 |       |                 |       |                 |       |
|--------|-------------|----------------------------------------------------------|-------------------|-------|-----------------|-------|-----------------|-------|-----------------|-------|
|        |             |                                                          | Part A            |       | Part B          |       | Part C          |       | Total           |       |
|        |             |                                                          | No of Questions   | Marks | No of Questions | Marks | No of Questions | Marks | No of Questions | Marks |
| 1      | 14          | 30                                                       | 3                 | 3     | 2               | 6     | 2               | 14    | 7               | 23    |
| 2      | 17          | 36                                                       | 2                 | 2     | 4               | 12    | 4               | 28    | 10              | 42    |
| 3      | 14          | 30                                                       | 2                 | 2     | 2               | 6     | 4               | 28    | 8               | 36    |
| 4      | 13          | 27                                                       | 2                 | 2     | 2               | 6     | 2               | 14    | 6               | 22    |

|       |    |     |   |   |    |    |    |    |    |     |
|-------|----|-----|---|---|----|----|----|----|----|-----|
| Total | 58 | 123 | 9 | 9 | 10 | 30 | 12 | 84 | 31 | 123 |
|-------|----|-----|---|---|----|----|----|----|----|-----|

**Blue Print**

**Cognitive Level Mark  
Distribution**

| <b>Cognitive Level</b> | <b>Marks</b> | <b>% of Marks</b> |
|------------------------|--------------|-------------------|
| Remembering            | 8            | 6                 |
| Understanding          | 56           | 46                |
| Applying               | 59           | 48                |
| Analysing              |              |                   |
| Evaluating             |              |                   |
| Creating               |              |                   |
| <b>Total</b>           | <b>123</b>   | <b>100</b>        |

**SCHEME OF VALUATION**  
**(Scoring Indicators)- Set II**  
**Quantitative Techniques**

| <b>Qn.<br/>No</b> | <b>Scoring Indicators</b> | <b>Split<br/>up<br/>Score</b> | <b>Sub<br/>Total</b> | <b>Total</b> |
|-------------------|---------------------------|-------------------------------|----------------------|--------------|
| <b>I</b>          | <b>PART A</b>             |                               |                      |              |
| 1                 | Primary data              | 1                             | 1                    | 1            |
| 2                 | Classification            | 1                             | 1                    | 1            |
| 3                 | Mode                      | 1                             | 1                    | 1            |
| 4                 | 85-10 = <b>75</b>         | 1                             | 1                    | 1            |
| 5                 | Rank Correlation          | 1                             | 1                    | 1            |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |   |   |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---|
| 6  | Negative Correlation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1   | 1 | 1 |
| 7  | Index Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1   | 1 | 1 |
| 8  | Primary data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1   | 1 | 1 |
| 9  | Fisher's Price Index                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1   | 1 | 1 |
| II | <b>PART B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |   |   |
| 1  | <b>Essentials of good tabulation</b> <ol style="list-style-type: none"> <li>1. Table should be simple, brief and neat.</li> <li>2. The tables should not be overloaded with details.</li> <li>3. The items should be arranged in alphabetical or chronological order.</li> <li>4. A proper title at the top of each table should be provided.</li> <li>5. Headings must be precise and self-explanatory.</li> <li>6. Abbreviations should be avoided.</li> <li>7. Figures should be rounded.</li> </ol> <p>(Any 3 points mentioned above)</p> | 3X1 | 3 | 3 |
| 2  | <b>Merits of Arithmetic Mean</b> <ol style="list-style-type: none"> <li>1. Arithmetic Mean is simple to understand.</li> <li>2. Arithmetic Mean is easy to calculate.</li> <li>3. Its value is always definite.</li> <li>4. It is based on all the items of a series.</li> <li>5. It is capable of further algebraic treatment.</li> <li>6. Arithmetic mean is stable. It is not affected by fluctuations of sampling.</li> </ol> <p>(Any 3 points mentioned above)</p>                                                                       | 3X1 | 3 | 3 |





|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |   |   |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---|
|   | <p>Ascending order</p> <p>X : 24, 38, 40, 45, 50, 56, 60, 63, 85, 95</p> <p><b>Median = Size of <math>\frac{N+1}{2}</math>th item.</b></p> <p>= Size of <math>\frac{10+1}{2}</math>th item</p> <p>= Size of <math>\frac{11}{2}</math>th item = Size of 5.5<sup>th</sup> item.</p> <p>Median = <math>\frac{\text{Size of 5<sup>th</sup> item} + \text{size of 6<sup>th</sup> item}}{2}</math></p> <p>= <math>\frac{50+56}{2}</math></p> <p>= <math>\frac{106}{2} = \underline{\underline{53}}</math></p> | 1   |   |   |
|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1   |   |   |
| 7 | <p><b>Importance of Correlation</b></p> <p>1.It is used in deriving the degree and direction of relationship between the variables.</p> <p>2.It is used to reducing the range of uncertainty</p> <p>3.In economics, it is used in understanding the economic behaviour.</p> <p>4.In business, it is used to estimate the cost of sales, sales price etc.</p> <p>5.In science, it is used in measuring the progress.</p> <p>(Any 3 points mentioned above)</p>                                           | 3X1 | 3 | 3 |
| 8 | <p><b>a. Simple aggregative Method:</b></p> <p>This is the simplest method of constructing index numbers. Under this method, the total of current year prices of all the</p>                                                                                                                                                                                                                                                                                                                            | 1.5 | 3 | 3 |



|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |   |   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---|
|     | $= (3 \times 50) - (2 \times 52)$ $= 150 - 104$ $= \underline{46}$                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |   |   |
| 10  | <p><b>Demerits of Median</b></p> <ol style="list-style-type: none"> <li>1. Median is not based on all the observations.</li> <li>2. It is not suitable for further mathematical treatment.</li> <li>3. For calculating median, it is necessary to arrange the data in ascending or descending order.</li> <li>4. Median is relatively less stable than median.</li> <li>5. Median being a positional average is not based on each and every item of the distribution.</li> </ol> <p>(Any 3 points mentioned above)</p> | 3X1 | 3 | 3 |
|     | <b>PART C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |   |   |
| III | <p><b>Methods of collecting Primary Data</b></p> <ol style="list-style-type: none"> <li>1. Direct Personal Investigation</li> <li>2. Indirect oral investigation</li> <li>3. Mailed Questionnaire Method</li> <li>4. Schedules sent through enumerators</li> <li>5. Through Local Correspondents:</li> </ol> <p><b>Methods of collecting Secondary Data</b></p> <ol style="list-style-type: none"> <li>1. Published Sources</li> <li>2. Unpublished Sources</li> </ol>                                                 | 7X1 | 7 | 7 |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |             |   |    |      |   |   |    |       |    |    |     |       |    |    |     |       |    |    |     |       |   |    |     |  |           |  |             |                   |   |   |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------|---|----|------|---|---|----|-------|----|----|-----|-------|----|----|-----|-------|----|----|-----|-------|---|----|-----|--|-----------|--|-------------|-------------------|---|---|
| 1V    | <p><b>Functions of Statistics</b></p> <p>1. It simplifies Complexity:</p> <p>2. It presents facts in a definite and precise form:</p> <p>3.It enables comparison of data:</p> <p>4. It helps in testing:</p> <p>5. It helps forecasting:</p> <p>6. It helps in formulating policies:</p> <p>7. Statistics studies the relationship between different groups:</p> <p>8. It enlarges human experience:</p> <p>(Any 7 points mentioned above)</p>                                                                                                                                                                        | 7X1 | 7           | 7 |    |      |   |   |    |       |    |    |     |       |    |    |     |       |    |    |     |       |   |    |     |  |           |  |             |                   |   |   |
| V     | <p><b>Computation of Mean</b></p> <table><tr><td>X</td><td>F</td><td>X</td><td>Fx</td></tr><tr><td>0-10</td><td>5</td><td>5</td><td>25</td></tr><tr><td>10-20</td><td>12</td><td>15</td><td>180</td></tr><tr><td>20-30</td><td>14</td><td>25</td><td>350</td></tr><tr><td>30-40</td><td>10</td><td>35</td><td>350</td></tr><tr><td>40-50</td><td>8</td><td>45</td><td>360</td></tr><tr><td></td><td><b>49</b></td><td></td><td><b>1265</b></td></tr></table> <p><b>Arithmetic Mean = <math>\frac{\sum fx}{N}</math></b></p> <p><b>N</b></p> <p><b>= <math>\frac{1265}{49}</math></b></p> <p><b>= <u>25.82</u></b></p> | X   | F           | X | Fx | 0-10 | 5 | 5 | 25 | 10-20 | 12 | 15 | 180 | 20-30 | 14 | 25 | 350 | 30-40 | 10 | 35 | 350 | 40-50 | 8 | 45 | 360 |  | <b>49</b> |  | <b>1265</b> | <p>4</p> <p>3</p> | 7 | 7 |
| X     | F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | X   | Fx          |   |    |      |   |   |    |       |    |    |     |       |    |    |     |       |    |    |     |       |   |    |     |  |           |  |             |                   |   |   |
| 0-10  | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5   | 25          |   |    |      |   |   |    |       |    |    |     |       |    |    |     |       |    |    |     |       |   |    |     |  |           |  |             |                   |   |   |
| 10-20 | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 15  | 180         |   |    |      |   |   |    |       |    |    |     |       |    |    |     |       |    |    |     |       |   |    |     |  |           |  |             |                   |   |   |
| 20-30 | 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 25  | 350         |   |    |      |   |   |    |       |    |    |     |       |    |    |     |       |    |    |     |       |   |    |     |  |           |  |             |                   |   |   |
| 30-40 | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 35  | 350         |   |    |      |   |   |    |       |    |    |     |       |    |    |     |       |    |    |     |       |   |    |     |  |           |  |             |                   |   |   |
| 40-50 | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 45  | 360         |   |    |      |   |   |    |       |    |    |     |       |    |    |     |       |    |    |     |       |   |    |     |  |           |  |             |                   |   |   |
|       | <b>49</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     | <b>1265</b> |   |    |      |   |   |    |       |    |    |     |       |    |    |     |       |    |    |     |       |   |    |     |  |           |  |             |                   |   |   |





|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |   |   |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---|
| VIII | <b>Types of Correlation</b><br><br><b>1. Simple, Partial and Multiple Correlation</b><br><br><p>When the relationship between any two variables only studied, it is a case of simple correlation.</p> <p>When the relationship between any two out of three or more variables is studied, ignoring the effect of the other related variables, it is a case of partial correlation.</p> <p>When the relationship between three or more variables is studied simultaneously, it is a case of multiple correlation.</p>                                                                                                                                                                                     | 7X1 | 7 | 7 |
|      | <b>2. Positive and Negative Correlation</b><br><br><p>A <b>positive correlation</b> is one where values of both the variables under study move in the same direction. i.e, with an increase in the values of one variable, the values of other variable increases and with a decrease in the values of one variable, the values of other variable also decreases.</p> <p>A <b>negative correlation</b> is one where both the variables under study move in the opposite direction. i.e, an increase in the values of one variable is followed by a decrease in the values of the other, and a decrease in the values of one variable is followed by an increase in the values of the other variable.</p> |     |   |   |
|      | <b>3. Linear and Non- linear Correlation</b><br><br><p>When the amount of change in one variable leads to a constant ratio of change in the other variable, correlation is said to be linear.</p> <p>Correlation is said to be non-linear, when the amount of change in one variable is not in constant ratio to the change in the other variable.</p>                                                                                                                                                                                                                                                                                                                                                   |     |   |   |

| Commodities | p0 | p1 | q0 | q1 | p1q0 | p0q0 | P1q1 | p0q1 |
|-------------|----|----|----|----|------|------|------|------|
| A           | 5  | 7  | 3  | 4  | 21   | 15   | 28   | 20   |
| B           | 8  | 12 | 7  | 5  | 84   | 56   | 60   | 40   |
| C           | 15 | 20 | 6  | 4  | 120  | 90   | 80   | 60   |
| D           | 6  | 8  | 3  | 4  | 24   | 18   | 32   | 24   |
|             |    |    |    |    | 249  | 179  | 200  | 144  |

$$P01(DB) = \frac{\sum p1q0}{\sum p0q0} + \frac{\sum p1q1}{\sum p0q1} \times 100$$

$$= \frac{249}{179} + \frac{200}{144} \times 100$$

$$= \frac{1.39}{2} + \frac{1.39}{2} \times 100$$

$$= \underline{139}$$



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |   |   |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---|
| X | <p><b>1. Weighted Aggregative Index Numbers:</b></p> <p>These indices are of the simple aggregative type with the difference that weights are assigned to the various items included in the index. Various methods of constructing index numbers have been evolved on the basis of different methods of assigning weights.</p> <p><b>a. Laspeyre's Method:</b></p> <p>This method is devised by German Economist Laspeyre in 1871 for calculating price index. Under the method, we assume that the quantities consumed in the base year and current year are same and weights are determined by the quantities in the base year.</p> $P_{01}(L) = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$ <p><b>b. Paasche's Method:</b></p> <p>German Mathematician Paasche has introduced this method in 1874 as improvement over the Laspeyre's method. Under this method, he has taken quantities of current year as the respective weights of items.</p> $P_{01}(P) = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100$ <p><b>c. Dorbish and Bowley Method:</b></p> <p>Under this method, Dorbish and Bowley have suggested for simple arithmetic mean of the Laspeyre's and Paasche's indexes as the price index.</p> $P_{01}(DB) = \frac{L+P}{2}$ <p><b>d. Fisher's Ideal Index Method:</b></p> <p>The method devised by Prof. Irving Fischer for construction of index number is popularly known as Fischer's Ideal Index. It is the geometric mean of the Laspeyre's and Paasche's Index numbers. The formula for the Fisher's Index Number is</p> $P_{01}(F) = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100$ <p><b>e. Marshall – Edgeworth Method:</b></p> <p>In this method also both the current year and the base year prices and quantities are considered.</p> | 7X1 | 7 | 7 |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---|

$$P01 (ME) = \frac{\Sigma p_1 q_0 + \Sigma p_1 q_1}{\Sigma p_0 q_0 + \Sigma p_0 q_1} \times 100$$

**f. Kelley's Method:**

Under this method, average of quantities of base year and current year is usually taken as weight. It can also be base year or current year quantity or any other fixed quantity.

$$P01 (K) = \frac{\Sigma p_1 q}{\Sigma p_0 q} \times 100$$

$$\text{Where } q = \frac{q_0 + q_1}{2}$$

**2. Weighted Average of relative Index Number**

In this method, appropriate weights are assigned to the commodities according to the relative importance of those commodities in the group. Thus the index for the whole group is obtained on taking the weighted averages of the price relatives.

| XI        | <b>Computation of Median</b> <table><tr><th>Class</th><th>F</th><th>Cf</th></tr><tr><td>10.5-15.5</td><td>7</td><td>7</td></tr><tr><td>15.5-20.5</td><td>10</td><td>17</td></tr><tr><td>20.5-25.5</td><td>13</td><td>30</td></tr><tr><td>25.5-30.5</td><td>26</td><td>56</td></tr><tr><td>30.5-35.5</td><td>35</td><td>91</td></tr><tr><td>35.5-40.5</td><td>22</td><td>113</td></tr><tr><td>40.5-45.5</td><td>11</td><td>124</td></tr><tr><td>45.5-50.5</td><td>5</td><td>129</td></tr><tr><td></td><td><b>N =129</b></td><td></td></tr></table> <b>Median = <math>L1 + \frac{(N/2 - cf)}{f} \times I</math></b><br><br>N/2 = 129/2 = 64.5,<br>Median class 30.5-35.5<br>f = 35<br>Cf = 56<br>i=5<br>Median = $30.5 + \frac{64.5 - 56}{35} \times 5$<br>= $30.5 + \frac{8.5}{7}$<br>= 30.5+1.2<br>= <b><u>31.7</u></b> | Class | F  | Cf | 10.5-15.5 | 7 | 7 | 15.5-20.5 | 10 | 17 | 20.5-25.5 | 13 | 30 | 25.5-30.5 | 26 | 56 | 30.5-35.5 | 35 | 91 | 35.5-40.5 | 22 | 113 | 40.5-45.5 | 11 | 124 | 45.5-50.5 | 5 | 129 |  | <b>N =129</b> |  | 4 | 7 | 7 |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|-----------|---|---|-----------|----|----|-----------|----|----|-----------|----|----|-----------|----|----|-----------|----|-----|-----------|----|-----|-----------|---|-----|--|---------------|--|---|---|---|
|           | Class                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | F     | Cf |    |           |   |   |           |    |    |           |    |    |           |    |    |           |    |    |           |    |     |           |    |     |           |   |     |  |               |  |   |   |   |
|           | 10.5-15.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 7     | 7  |    |           |   |   |           |    |    |           |    |    |           |    |    |           |    |    |           |    |     |           |    |     |           |   |     |  |               |  |   |   |   |
| 15.5-20.5 | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 17    |    |    |           |   |   |           |    |    |           |    |    |           |    |    |           |    |    |           |    |     |           |    |     |           |   |     |  |               |  |   |   |   |
| 20.5-25.5 | 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 30    |    |    |           |   |   |           |    |    |           |    |    |           |    |    |           |    |    |           |    |     |           |    |     |           |   |     |  |               |  |   |   |   |
| 25.5-30.5 | 26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 56    |    |    |           |   |   |           |    |    |           |    |    |           |    |    |           |    |    |           |    |     |           |    |     |           |   |     |  |               |  |   |   |   |
| 30.5-35.5 | 35                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 91    |    |    |           |   |   |           |    |    |           |    |    |           |    |    |           |    |    |           |    |     |           |    |     |           |   |     |  |               |  |   |   |   |
| 35.5-40.5 | 22                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 113   |    |    |           |   |   |           |    |    |           |    |    |           |    |    |           |    |    |           |    |     |           |    |     |           |   |     |  |               |  |   |   |   |
| 40.5-45.5 | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 124   |    |    |           |   |   |           |    |    |           |    |    |           |    |    |           |    |    |           |    |     |           |    |     |           |   |     |  |               |  |   |   |   |
| 45.5-50.5 | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 129   |    |    |           |   |   |           |    |    |           |    |    |           |    |    |           |    |    |           |    |     |           |    |     |           |   |     |  |               |  |   |   |   |
|           | <b>N =129</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |    |    |           |   |   |           |    |    |           |    |    |           |    |    |           |    |    |           |    |     |           |    |     |           |   |     |  |               |  |   |   |   |
| 3         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |    |    |           |   |   |           |    |    |           |    |    |           |    |    |           |    |    |           |    |     |           |    |     |           |   |     |  |               |  |   |   |   |
|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |    |    |           |   |   |           |    |    |           |    |    |           |    |    |           |    |    |           |    |     |           |    |     |           |   |     |  |               |  |   |   |   |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |     |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|---|
| XII | <b>Measures of Central Tendency</b> means the methods of finding out the central value or average value of a statistical series. An average is a single figure that is computed from a given series to give a central value about the central series.<br><br><b>Importance</b><br><br>1. Averages give a general idea about the whole group:<br><br>2. Averages can be used for summarizing the data:<br><br>3. Averages help comparison:<br><br>4. Averages help in decision making:<br><br>5. Averages constitute the basis of statistical analysis: | 1 | 1+6 | 7 |
|     | 6X1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |     |   |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |     |   |

|      |                                                        |    |            |                |   |   |   |
|------|--------------------------------------------------------|----|------------|----------------|---|---|---|
|      | 6. Averages represents the universe:                   |    |            |                |   |   |   |
| XIII | R1                                                     | R2 | D ( R1-R2) | D <sup>2</sup> | 3 |   |   |
|      | 1                                                      | 2  | 1          | 1              |   |   |   |
|      | 5                                                      | 1  | 4          | 16             |   |   |   |
|      | 3                                                      | 3  | 0          | 0              |   |   |   |
|      | 2                                                      | 5  | 3          | 9              |   |   |   |
|      | 10                                                     | 8  | 2          | 4              |   |   |   |
|      | 7                                                      | 7  | 0          | 0              |   |   |   |
|      | 4                                                      | 4  | 0          | 0              |   |   |   |
|      | 8                                                      | 6  | 2          | 4              |   |   |   |
|      | 9                                                      | 10 | 1          | 1              |   | 7 | 7 |
|      | 6                                                      | 9  | 3          | <u>9</u>       | 4 |   |   |
|      | <b>44</b>                                              |    |            |                |   |   |   |
|      | <b>R = 1 - <math>\frac{6 \sum D^2}{N^3 - N}</math></b> |    |            |                |   |   |   |
|      | = 1 - $\frac{6 \times 44}{1000 - 10}$                  |    |            |                |   |   |   |
|      | = 1 - $\frac{264}{990}$                                |    |            |                |   |   |   |
|      | = 1 - .267 = <b><u>0.733</u></b>                       |    |            |                |   |   |   |

